Condition Material Generation

August 24, 2026

1 ToDo List

- Improving performance for Equiformer with Mattergen(Dipannoy)
  - Try lower(higher) hidden dimension
  - add lattice to the node embeddings
  - investigating output dimension of lattice
- ~~Merging Table 1 and Table 2 (Chris)~~
- ~~Fill in table Table 1 with estimated values (Chris)~~
- ~~Tune hyperparameters for CrystalGRW for AABBX6 (Chote)~~
- Calculate final run of results from CrystalGRW for AABBX6 (Chris)
- Tune hyperparameters for CrystalGRW for MP (Chote, Nima)
- Writing manuscript (All)

2 Summary of main conclusions

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3 Introduction

The generation of new materials has been an long time research topic for the scientists. The generative models have accelerated this process by many folds. This discovery of new materials can lead to a groundbreaking success for many technologies. Conditional material generation helps to generate materials with specific properties. This kind of materials can come into use for specific applications.

Inverse design for materials is a cutting-edge approach in materials science that flips the conventional design process on its head by starting with target properties and letting AI generate candidate materials. Inverse design refers to designing materials by starting with the desired properties (like bandgap, stability, formation energy) and using generative machine learning models to directly produce candidate structures that satisfy them. It contrasts with forward methods, where many candidate materials are screened to find ones that meet the criteria.

Band gap is a fundamental property for inorganic crystals which governs the electronic and optical behavior. The electronic classification determines whether a crystal is a metal, semi-conductor, or insulator. The optical properties dictate absorption/emission wavelength, important for LEDs, lasers, and photovoltaics. As a result, modeling and predicting the band gap is considered a vital task in material discovery. However, the prediction of band gap remains a challenging task for the machine learning models even the DFT datasets often underestimate experimental band gaps. Generative models also face difficulties to generate materials with specific bangap.

The Advancement in Generative Models is showing a promising path to explore new materials specifically inorganic crystals. A good number of generative models have been developed based on the different theory incorporating forces, energy, and atomic environments. All models aim to generate Novel, stable materials. However, generation with guidance can make the new materials more useful. Therefore, the generative models are trying to learn the different properties. This learning can be used later during generation. The properties are injected differently across the generative models.

DiffCSP works with crystal-graph representation following an iterative refinement process. The objective is to generate physically plausible materials through the refinement process using a diffusion-based framework. The conditioning is done by emdedding the target property into latent vector. The vector is concatinated with the input at each step of diffusion process. This guides the model to generate crystal structures aligned with the desired property.

CDVAE (Crystal Diffusion Variational Autoencoder) is a generative model that combines a VAE with a diffusion process to generate crystal structures. It learns a smooth latent space of materials and uses a denoising diffusion mechanism for high-quality structure generation. CDVAE enables conditional generation based on desired properties while maintaining crystallographic symmetry and validity. The conditioning is performed by projecting target properties into the latent space of the VAE. During training and sampling, this conditioned latent vector guides the diffusion decoder to generate crystal structures with the desired properties.

Con-CDVAE (Conditioned Crystal Diffusion Variational Autoencoder) is an extension of CDVAE designed for property-guided crystal structure generation. The key difference in conditioning between CDVAE and Con-CDVAE lies in how deeply and consistently the property information is integrated. CDVAE conditions only at the sampling stage by projecting the target property into the latent space, guiding the diffusion decoder after the VAE has encoded the structure. Con-CDVAE, in contrast, introduces conditioning during both training and sampling. It conditions both the encoder and decoder, ensuring the model learns a joint representation of crystal structures and properties. This leads to stronger and more consistent alignment between the generated structures and the desired properties.

In DP-CDVAE (Diffusion-Posterior CDVAE), conditioning is performed by integrating target properties into the posterior distribution learned by the VAE. The model encodes both structure and property into a joint latent space during training. At generation time, the property-conditioned latent vector guides the diffusion decoder to produce crystals with desired properties.

FTCP is an another framework for general inverse design of inorganic crystals consists of two components. First component creates the crystallographic representation, which is both composition and structure varying. The second component is the combination of the encoder and decoder with an extra target-learning branch connected to the latent space, mapping latent vectors/points to user-specified property(ies).

4 Related Works

Szymanski & Bartel (2025)[3] worked with two traditional material generation approaches to compare their performance against four generative models: CrystaLLM, FTCP, CDVAE, and MatterGen. The evaluation has been performed considering three objectives: stability and novelty of materials, materials with a bandgap of 3 eV, and materials with high bulk modulus.

LeMat-GenBench[1] introduces a uniform benchmark work, followed by a set of evaluation schemes to facilitate the model development and downstream prediction tasks. They have completed their benchmarking across 12 generative models, where not a single model can perform better for all metrics. The authors have defined three benchmark scenarios: unconditional generation, conditional generation, and conditional generation with limited budget, considering three fundamental tasks in the generation. The framework shows a detailed evaluation of stability, uniqueness, and novelty and their mutual comparison to highlight the tradeoff of different metrices.

Phononbench[2] addressed the problem of dynamical stability of the AI generated crystals. They have used Mattersim interatomic potentials to evaluate 108843 crystal structures generated by six

crystal generation models. The results depict the limitation of current generative models in ensuring phonon stability where average dynamical stability has been found only 25.83% across all models. This paper has tried to demonstrate thermodynamic metrics are alone not sufficient to evaluate the performnce of generative models.

5 Methods

In this work, we have worked with MatterGen and CrystalGRW to examine the conditioning scheme across two different decoder graph neural networks.

MatterGen is a generative framework designed for crystal structure generation with explicit property control. The model builds on recent advances in equivariant graph neural networks and diffusion-based generative modeling.

A central feature of MatterGen is its conditioning mechanism. Instead of generating arbitrary crystals, the model incorporates external constraints—such as target material properties (e.g., band gap, formation energy, stability indicators), composition, or symmetry. Conditioning is achieved by injecting the property embeddings into the generative process, either through feature concatenation at each layer or via cross-attention with the latent representations of atomic graphs. This allows MatterGen to generate candidate materials that not only obey crystallographic constraints but are also biased toward desired physical properties.

The training is performed in two stages. The training without conditioning scheme is called the pre-trainng while the training with conditioning scheme is performed with an additional adapter module where the pretrained model is used to initialize the previous layers. This process is also called finetuning.

The performance of Mattergen has also been investigated without adapter module.

For the decoder, MatterGen employs GemNet, an advanced message-passing neural network originally developed for predicting molecular and crystal properties. GemNet is well-suited as a decoder because it captures both geometric (bond lengths, angles, torsions) and topological (connectivity) features of atomic structures with high accuracy. By leveraging equivariant message passing over atoms and edges, GemNet ensures that the generated atomic positions and lattice vectors remain consistent with the underlying physics of crystals. In MatterGen, GemNet acts as a structure–property bridge, refining intermediate representations from the diffusion model into realistic atomic configurations while respecting the conditioning signals.

Together, the conditioning module and GemNet-based decoder enable MatterGen to move beyond unconstrained generative models, offering a controllable framework for inverse materials design: proposing Novel crystals that are not only physically valid but also optimized for user-specified properties.

5.1 More Target Hits with Skewed Dataset

Maximum log likelihood estimation (MLE) aims to estimate the parameters of an assumed probability distribution, given some observed data. This is achieved by maximizing a likelihood function so that, under the assumed statistical model, the observed data is most probable.

In generative modeling, the base objective is maximum likelihood estimation (MLE). Given a data distribution $p_{\text{data}}(x)$ and model $p_\theta(x)$, the objective is

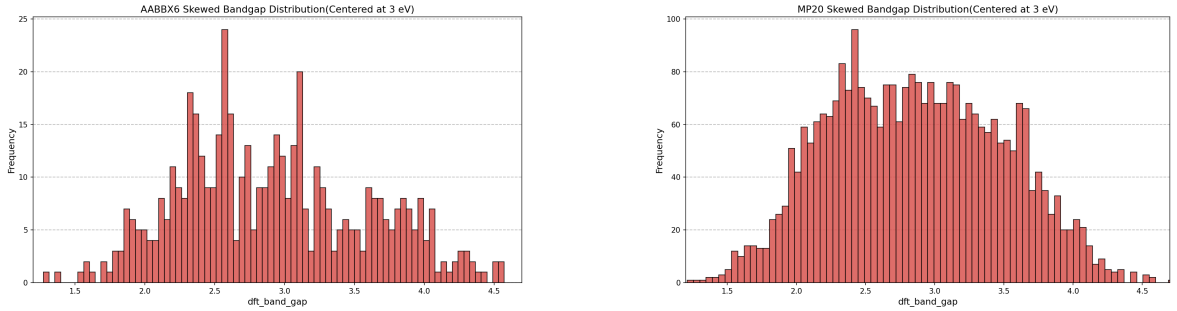
$$\mathcal{L}_{\text{MLE}}(\theta) = \mathbb{E}_{x\sim p_{\text{data}}}\big[\log p_\theta(x)\big]. \tag{1}$$

Here, the model $p_\theta(x)$ is the θ parametrized. The parameter is updated based on the data distribution $p_{\text{data}}(x)$. MLE guides the model to follow the given data distribution.

For diffusion models (as used in MatterGen), this corresponds to denoising score matching:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{t,x_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta(x_t, t, c)\right\|^2\right], \tag{2}$$

where x_t is the noisy sample at timestep t, ϵ is the Gaussian noise added, c is the conditioning variable (e.g., desired material property), ϵ_θ is the neural network predicting the noise.



(a) (b)

Figure 1: (a)Distribution of DFT Bandgap for the Skewed AABBX6 Dataset, (b)Distribution of DFT Bandgap for the Skewed MP20 Dataset,

In the denoising process, the model is trained to predict the noise to produce data having the distribution of the original data. So the objective of this function is to maximize the likelihood of the noisy data towards the original data. This denoising score function is extended with two different calculations while the condition is incorporated.

$$s_\theta(x_t, t, c) = \nabla_{x_t} \log p_\theta(x_t \mid c) \tag{3}$$

Here, the model learns both from the noisy data x_t and the condition c. x_t denotes the noisy state of the data at time step t, the gradient of *log* drives the noisy data towards the original distribution. As denoising score function takes care both of the conditional score and unconditional score, context-free guidance is performed.

$$\nabla_{x_t} \log p_\theta(x_t \mid c)_{\text{guided}} \;\approx\; (1-w)\,\nabla_{x_t} \log p_\theta(x_t) \;+\; w\,\nabla_{x_t} \log p_\theta(x_t \mid c) \tag{4}$$

Here, $\nabla_{x_t} \log p_\theta(x_t)$ is the unconditional score and $\nabla_{x_t} \log p_\theta(x_t \mid c)$ is the conditional score, w is the guided strength. This scoring is considered during sampling. Unconditional training aims to learn the data distribution by maximizing the likelihood of noisy data towards the original data. It ensures that the newly generated data will be realistic and close to the ground truth. On the other hand, the conditional training makes the model learn about the property by integrating the property at node level.

5.2 Skewed Dataset Training

We have investigated the inverse design with a specific target property with a fine-tuning scheme skewing the total dataset following a distribution centered at 3 eV. From the AABBX6 dataset, such 500 training samples have been extracted where for the MP20 dataset we have taken 3000 samples to fine-tune the model.

6 FTCP Higher Percentage in Target Bandgap Generation

From Chris Bartel paper, FTCP is found as the best model to generate the higher percentage(68.8%) of crystals with the bandgap 2.5 to 3.5 eV. However, a large number of generated structures come from the existing MP 20 dataset. FTCP generates total 498 structures where 333 structures have been found in the current database leading to 165 novel structures. Out of 165 structures, 117 structures have been tested valid in both composition and structure.

We have also examined the bandgap range for the generated structures already found in the database. From the total generation, 137 old structures have been found with the badgap from 2.5 to 3.5 eV. Those structures also contributed to the higher percentage reported in the paper. In case of valid novel structures, if all 117 structures show the bandgap in the target region the percentage

comes down to less than 30%.

We suspect VAE model has the tendency to generate structures which are already in the training set which causes the higher performance in target bandgap generation. Therefore, FTCP yields a high performance in conditional control because it replicates the training set rather than generated new structures. If valid new strcutres are taken into account the, percentage will drop to less than 30% from 68.8%.

7 Results

Table 1: Comparative analysis for the 600 **10 atom** structures generated from different generative models using different conditioning schemes with different GNNs as decoder. Models were fine-tuned with AABBX6. (Cyan texts are tasks got generated. Green texts are tasks being training.)

Model	GNN	Adapter	PT (pre-train)	FT (fine-tune)	Valid	Novel	Comps in database	Estimated Eg = 3±0.5 eV	DFT Eg = 3±0.5 eV (%)	DFT converged
Mattergen	Equiformer	On	ALEXMP20	AABBX6	546	356	190	3.16%		
				AABBX6(skew)	452	303	149	9.4		
			AABBX6	AABBX6	598	259	339	2.95%		
				AABBX6(skew)	596	211	385	10.65%		
		Off		AABBX6	594	262	332	2.71%	-	-
				AABBX6(skew)	401	238	163	20.86%	-	-
	GemNet	On	ALEXMP20	AABBX6	565	221	345	9.28%	13.2%	529
				AABBX6(skew)	539	239	300	16.33%	29%	529
			AABBX6	AABBX6	591	197	394	10.15%	19.5%	536
				AABBX6(skew)	591	172	419	17.66%		
		Off		AABBX6	293	293	0	0		
				AABBX6(skew)	Generation unsuccessful					
CrystalGRW	Equiformer	On	ALEXMP20	AABBX6	—	—	—	—	—	—
				AABBX6(skew)	—	—	—	—	—	—
			AABBX6	AABBX6	548	—	535	5.56%(old)	4.5%(old)	532
				AABBX6(skew)	—	—	—	—	—	—
		Off		AABBX6	476	246	230	8.26%	10.2%	445
				AABBX6(skew)	65	65	0	0		
	GemNet	On	ALEXMP20	AABBX6	—	—	—	—	—	—
				AABBX6(skew)	—	—	—	—	—	—
			AABBX6	AABBX6	549	—	391	3.83%(old)	6.5%(old)	246
				AABBX6(skew)	—	—	—	—	—	—
		Off		AABBX6	464	239	225	3.11%	13.0%	454
				AABBX6(skew)	206	180	26	0		

Table 2: Comparative analysis for the 600 structures with **2–20 atoms** generated from different generative models using different conditioning schemes with different GNNs as decoder. For the adapter module, the models are pretrained with the **ALEX-MP-20** dataset. (Cyan texts are tasks got generated. Green texts are tasks being training. Red texts are not yet working on.)

Model	GNN	Adapter	FT (fine-tune)	Valid	Novel	Comps in database	Estimated Eg = 3±0.5 eV (%)	DFT Eg = 3±0.5 eV (%)	DFT Converged
CrystalGRW	GemNet	Off	MP20	421	383	38	13.16%	—	—
		Off	MP20(skew)	455	447	8	37.5%	—	—
		On	~~MP20~~	—	—	—	—	—	—
		On	~~MP20(skew)~~	—	—	—	—	—	—
	EqV2	Off	MP20	127	124	3	0%	—	—
		Off	MP20(skew)	113	112	1	0%	—	—
		On	MP20	—	—	—	—	—	—
		On	MP20(skew)	—	—	—	—	—	—
Mattergen	GemNet	Off	MP20	514	405	109	72%	—	—
		Off	MP20(skew)	513	403	110	68%	—	—
		On	MP20	543	350	193	44%	—	—
		On	MP20(skew)	549	341	208	56%	—	—
	EqV2	Off	MP20	400	357	43	6.98%	—	—
		Off	MP20(skew)	389	380	9	44.44%	—	—
		On	MP20	334	314	20	0%	—	—
		On	MP20(skew)	304	283	21	42.83%		—
FTCP (Chris' B)	—	—	MP20	429	165	—	—	—	—

References

[1] Siddharth Betala, Samuel P Gleason, Ali Ramlaoui, Andy Xu, Georgia Channing, Daniel Levy, Clémentine Fourrier, Nikita Kazeev, Chaitanya K Joshi, Sékou-Oumar Kaba, et al. Lematgenbench: A unified evaluation framework for crystal generative models. arXiv preprint arXiv:2512.04562, 2025.

[2] Xiao-Qi Han, Peng-Jie Guo, Ze-Feng Gao, and Zhong-Yi Lu. Phononbench: A large-scale phonon-based benchmark for dynamical stability in crystal generation. arXiv preprint arXiv:2512.21227, 2025.

[3] Nathan J Szymanski and Christopher J Bartel. Establishing baselines for generative discovery of inorganic crystals. Materials Horizons, 12(19):8000–8011, 2025.

Table 3: Latest version of the code. CrystalGRW fine-tuned with AABBX6 dataset. Total gen is the number of total generated structures. Compos in database is the number of gen structures found in database. Btw 2.5-3.5 eV is the number of gen structures that have $2.5 <= E_g <= 3.5$ eV in database. LR is a learning rate, DO is a dropout in percentage, and GS is a guidance strength.

GNN	PT	Adapter	FT	LR	GS	Total gen (x)	Compos in database (y)	Btw 2.5-3.5 eV (z)	z/x (%)	z/y (%)	DFT (%)
EqV2 (128)	AlexMP20	MLP	–	10^{-5}	0.0	640	167	11	1.72	6.59	
			–	10^{-5}	0.5	640	149	10	1.56	6.71	
			–	10^{-6}	0.0	640	184	7	1.09	3.8	
			–	10^{-6}	0.5	640	186	8	1.25	4.3	
			skew	10^{-5}	0.0	640	161	7	1.09	4.35	
			skew	10^{-5}	0.5	640	170	10	1.56	5.88	
			skew	10^{-6}	0.0	640	88	9	1.41	10.23	
			skew	10^{-6}	0.5	640	94	17	2.66	18.09	
		Fourier	–	10^{-5}	0.0	640	168	9	1.41	5.36	
			–	10^{-5}	0.5	640	157	12	1.88	7.64	
			–	10^{-6}	0.0	640	169	3	0.47	1.78	
			–	10^{-6}	0.5	640	164	7	1.09	4.27	
			skew	10^{-5}	0.0	640	123	6	0.98	4.88	
			skew	10^{-5}	0.5	640	130	10	1.56	7.69	
			skew	10^{-6}	0.0	640	104	11	1.72	10.58	
			skew	10^{-6}	0.5	640	106	10	1.56	9.43	
	AABBX6	MLP	–	10^{-5}	0.0	640	309	32	5.0	10.36	
			–	10^{-5}	0.5	640	297	36	5.62	12.12	
			–	10^{-6}	0.0	640	304	39	6.09	12.83	
			–	10^{-6}	0.5	640	293	29	4.53	9.9	
			skew	10^{-5}	0.0	640	306	25	3.91	8.17	
			skew	10^{-5}	0.5	640	317	36	5.62	11.36	
			skew	10^{-6}	0.0	640	337	33	5.16	9.97	
			skew	10^{-6}	0.5	640	316	38	5.94	12.03	
		Fourier	–	10^{-5}	0.0	640	289	23	3.59	7.96	
			–	10^{-5}	0.5	640	311	32	5.0	10.29	
			–	10^{-6}	0.0	640	301	20	3.12	6.64	
			–	10^{-6}	0.5	640	303	25	3.91	8.25	
			skew	10^{-5}	0.0	640	316	33	5.16	10.44	
			skew	10^{-5}	0.5	640	320	35	5.47	10.94	
			skew	10^{-6}	0.0	640	324	40	6.25	12.35	
			skew	10^{-6}	0.5	640	287	30	4.69	10.45	
EqV2 (256)	AABBX6	MLP	–	10^{-5}	0.0	640	308	21	3.28	6.82	
			–	10^{-5}	0.5	640	306	29	4.53	9.48	
			–	10^{-6}	0.0	640	308	25	3.91	8.22	
			–	10^{-6}	0.5	640	300	37	5.78	12.33	
			skew	10^{-5}	0.0	640	63	7	1.09	11.11	
			skew	10^{-5}	0.5	640	77	11	1.72	14.29	
			skew	10^{-6}	0.0	640	86	7	1.09	8.14	
			skew	10^{-6}	0.5	640	68	5	0.78	7.35	
		Fourier	–	10^{-5}	0.0	640	286	33	5.16	11.54	
			–	10^{-5}	0.5	640	335	28	4.38	8.36	
			–	10^{-6}	0.0	640	280	26	4.06	9.26	
			–	10^{-6}	0.5	640	263	32	5.0	12.17	
			skew	10^{-5}	0.0	640	114	27	4.22	23.68	
			skew	10^{-5}	0.5	640	112	21	3.28	18.75	
			skew	10^{-6}	0.0	640	50	6	0.94	12.0	
			skew	10^{-6}	0.5	640	65	11	1.72	11.69	
GemNet (512) (4 blocks)	AlexMP20	MLP	–	10^{-5}	0.0	640					
			–	10^{-5}	0.5	640					
			–	10^{-6}	0.0	640					
			–	10^{-6}	0.5	640					
			skew	10^{-5}	0.0	640					
			skew	10^{-5}	0.5	640					
			skew	10^{-6}	0.0	640					
			skew	10^{-6}	0.5	640					
		Fourier	–	10^{-5}	0.0	640					
			–	10^{-5}	0.5	640					
			–	10^{-6}	0.0	640					
			–	10^{-6}	0.5	640					
			skew	10^{-5}	0.0	640					
			skew	10^{-5}	0.5	640					
			skew	10^{-6}	0.0	640					
			skew	10^{-6}	0.5	640					
	AABBX6	MLP	–	10^{-5}	0.0	640	194	14	2.19	7.22	
			–	10^{-5}	0.5	640	185	19	2.97	10.27	
			–	10^{-6}	0.0	640	210	22	3.44	10.48	
			–	10^{-6}	0.5	640	206	24	3.75	11.65	
			skew	10^{-5}	0.0	640	281	16	2.5	5.69	
			skew	10^{-5}	0.5	640	279	25	3.91	8.96	
			skew	10^{-6}	0.0	640	295	21	3.28	7.12	
			skew	10^{-6}	0.5	640	304	20	3.12	6.58	
		Fourier	–	10^{-5}	0.0	640	212	13	2.03	6.13	
			–	10^{-5}	0.5	640	175	12	1.88	6.86	
			–	10^{-6}	0.0	640	191	21	3.28	10.99	
			–	10^{-6}	0.5	640	146	13	2.03	8.9	
			skew	10^{-5}	0.0	640	286	34	5.31	11.89	
			skew	10^{-5}	0.5	640	289	24	3.75	8.3	
			skew	10^{-6}	0.0	640	337	27	4.22	8.01	
			skew	10^{-6}	0.5	640	310	18	2.81	5.81	

Table 4: Latest version of the code. CrystalGRW fine-tuned with AABBX6 dataset. Total gen is the number of total generated structures. Compos in database is the number of gen structures found in database. Btw 2.5-3.5 eV is the number of gen structures that have $2.5 <= E_g <= 3.5$ eV in database. LR is a learning rate, DO is a dropout in percentage, and GS is a guidance strength.

GNN	PT	Adapter	FT	LR	GS	Total gen (x)	Compos in database (y)	Btw 2.5-3.5 eV (z)	z/x (%)	z/y (%)	DFT (%)
GemNet (256) (4 blocks)	AABBX6	MLP	–	10^{-5}	0.0	640	292	22	3.44	7.53	
			–	10^{-5}	0.5	640	281	27	4.22	9.61	
			–	10^{-6}	0.0	640	259	20	3.12	7.72	
			–	10^{-6}	0.5	640	246	20	3.12	8.13	
			skew	10^{-5}	0.0	640	248	19	2.97	7.66	
			skew	10^{-5}	0.5	640	242	20	3.12	8.26	
			skew	10^{-6}	0.0	640	276	28	4.38	10.14	
			skew	10^{-6}	0.5	640	256	19	2.97	7.42	
		Fourier	–	10^{-5}	0.0	640	272	15	2.34	5.51	
			–	10^{-5}	0.5	640	309	31	4.84	10.03	
			–	10^{-6}	0.0	640	282	21	3.28	7.45	
			–	10^{-6}	0.5	640	261	18	2.81	6.9	
			skew	10^{-5}	0.0	640	301	21	3.28	6.98	
			skew	10^{-5}	0.5	640	305	20	3.12	6.56	
			skew	10^{-6}	0.0	640	311	23	3.59	7.4	
			skew	10^{-6}	0.5	640	302	25	3.91	8.28	

Table 5: Latest version of the code. CrystalGRW fine-tuned with MP20 dataset. Total gen is the number of total generated structures. LR is a learning rate, DO is a dropout in percentage, and GS is a guidance strength. Novelty is evaluated with the training set of the fine-tuning set. Non-novel ($E_g = 2.5 - 3.5$ eV) is the number of non-novel structures that have $2.5 <= E_g <= 3.5$ eV divided by the total number of non-novel structures.

GNN	PT	Adapter	FT	LR	GS	Total gen	Comp validity	Novelty	Valid&Novel	Non-novel ($E_g = 2.5 - 3.5$ eV)	DFT (%)
Eqv2 (Incorrect) (10 atoms)	AlexMP20	MLP	MP20	10^{-5}	0.0	640	0.8203	0.7953	0.6344	0.2977	
			MP20	10^{-5}	0.5	640	0.8141	0.8016	0.6359	0.2205	
			MP20_skew	10^{-6}	0.0	640	0.8047	0.8484	0.6531	0.7629	
			MP20_skew	10^{-6}	0.5	640	0.8250	0.8562	0.6813	0.7717	
Eqv2 (random) (1-20 atoms)	AlexMP20	MLP	MP20	10^{-5}	0.0	640	0.8641	0.7469	0.6219	0.0185	
			MP20	10^{-5}	0.5	640	0.8844	0.7516	0.6531	0.0126	
			MP20_skew	10^{-6}	0.0	640	0.9094	0.9719	0.8813	0.5556	
			MP20_skew	10^{-6}	0.5	640	0.8891	0.9781	0.8672	0.4286	
Eqv2 (random) (2-20 atoms)	AlexMP20	MLP	MP20	10^{-5}	0.0	640					
			MP20	10^{-5}	0.5	640					
			MP20_skew	10^{-6}	0.0	640					
			MP20_skew	10^{-6}	0.5	640					
Eqv2 (mp20 stat) (num atoms)	AlexMP20	MLP	MP20	10^{-5}	0.0	640					
			MP20	10^{-5}	0.5	640	0.8922	0.7312	0.6297	0.0523	
			MP20_skew	10^{-6}	0.0	640	0.9313	0.9734	0.9047	0.7059	
			MP20_skew	10^{-6}	0.5	640	0.9203	0.9688	0.8891	0.6000	